

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 01

Total Marks:100

Annexure A

CASE STUDY

Code of Conduct:

I Shaik Mohammed Abrar (192472185) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shaik Mohammed Abrar

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Levels	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly		
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts		
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning		
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided		
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand		

Annexure A

Case Study 7: Retail Sales Performance Analysis

Scenario:

A retail chain has collected sales data from different branches, including daily sales amount, number of customers, discount percentage, product category, and customer feedback score. The company wants to evaluate branch performance and improve sales planning.

Questions:

- How can descriptive statistics and distribution analysis be used to summarize daily sales and customer count?
- Discuss how abnormal sales values, such as festival-season sales or very low sales days, can affect the analysis. Explain suitable outlier detection methods.
- How can confidence intervals and hypothesis testing be used to determine whether the average sales of urban branches are significantly higher than rural branches?

Case Study 7: Retail Sales Performance Analysis

1. Introduction

Retail businesses generate large amounts of sales data from different branches. Analyzing variables such as daily sales, customer count, discount percentage, product category, and customer feedback helps management understand branch performance and improve sales planning. In this case study, statistical techniques such as descriptive statistics, distribution analysis, outlier detection, confidence intervals, and hypothesis testing are applied to compare the performance of urban and rural branches.

2. Problem Statement

The retail chain needs to analyze its sales data to:

1. Summarize daily sales and customer counts using descriptive statistics.
2. Understand the distribution of sales and customer activity.
3. Identify abnormal sales values caused by festival seasons or unusually low-sales days.
4. Determine whether urban branches have significantly higher average sales than rural branches.
5. Use statistical evidence to support better sales planning and branch-level decisions.

3. Objectives

1. Calculate descriptive statistics for daily sales and customer count.
2. Analyze the distribution of sales and customer counts.
3. Detect abnormal observations using statistical outlier detection methods.
4. Calculate confidence intervals for average sales.
5. Perform hypothesis testing between urban and rural branches.
6. Determine whether urban branches have significantly higher average sales.

4. Dataset Description

The generated dataset contains **120 retail sales records**.

Column	Description
Date	Date of sales record
Branch_Type	Urban or Rural
Daily_Sales	Daily sales amount
Customer_Count	Number of customers
Discount_Percentage	Discount offered
Product_Category	Category of products sold
Feedback_Score	Customer feedback score

Dataset

Retail Sales CSV

The dataset also contains a few intentionally abnormal sales values to demonstrate outlier detection.

Overall Conclusion

Question A: The average daily sales are **₹48,100.33**, with a standard deviation of **₹15,155.79**. The average daily customer count is **150.29**, with a standard deviation of **48.34**. The difference between mean and median sales indicates that unusually high sales influence the distribution.

Question B: The IQR method detects **three outliers**: ₹125,000, ₹115,000, and ₹5,000. The two very high values can represent genuine festival-season sales. Therefore, they should be investigated rather than automatically deleted.

Question C: Urban branches have an average sales value of **₹54,059.87**, compared with **₹39,756.96** for rural branches. The Welch t-test gives:

$t=6.313$, $p=2.98 \times 10^{-9}$

Since: $p < 0.05$ we reject the null hypothesis and conclude that urban branches have statistically significantly higher average sales than rural branches.

Code:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from scipy import stats

# -----

# 1. Load Dataset

# -----

df = pd.read_csv("retail_sales_case_study_7.csv")

print("\nRETAIL SALES DATA\n")

print(df.head())

print("\nDATA INFORMATION\n")

print(df.info())

# -----

# 2. Descriptive Statistics

# -----

print("\nDESCRIPTIVE STATISTICS\n")
```

```
print(df[["Daily\_Sales", "Customer\_Count"]].describe())
```

```
print("\nMean Sales:")
```

```
print(df["Daily\_Sales"].mean())
```

```
print("\nMedian Sales:")
```

```
print(df["Daily\_Sales"].median())
```

```
print("\nStandard Deviation of Sales:")
```

```
print(df["Daily\_Sales"].std())
```

```
print("\nMean Customer Count:")
```

```
print(df["Customer\_Count"].mean())
```

```
print("\nMedian Customer Count:")
```

```
print(df["Customer\_Count"].median())
```

```
print("\nStandard Deviation of Customer Count:")
```

```
print(df["Customer\_Count"].std())
```

```
# -----
```

```
# 3. Distribution Analysis
```

```
# -----
```

```
plt.figure(figsize=(8, 5))
```

```
sns.histplot(df["Daily\_Sales"], bins=20, kde=True)
```

```
plt.title("Distribution of Daily Sales")
```

```
plt.xlabel("Daily Sales")
```

```
plt.ylabel("Frequency")
```

```
plt.show()

plt.figure(figsize=(8, 5))

sns.histplot(df["Customer\_Count"], bins=20, kde=True)

plt.title("Distribution of Customer Count")

plt.xlabel("Customer Count")

plt.ylabel("Frequency")

plt.show()
```

```
# -----
```

```
# 4. Boxplot for Sales
```

```
# -----
```

```
plt.figure(figsize=(8, 5))

sns.boxplot(x=df["Daily\_Sales"])

plt.title("Boxplot of Daily Sales")

plt.xlabel("Daily Sales")

plt.show()
```

```
# -----
```

```
# 5. IQR Outlier Detection
```

```
# -----
```

```
Q1 = df["Daily\_Sales"].quantile(0.25)
```

```
Q3 = df["Daily\_Sales"].quantile(0.75)
```

```

IQR = Q3 - Q1

lower\_limit = Q1 - 1.5 \* IQR

upper\_limit = Q3 + 1.5 \* IQR

print("\nIQR OUTLIER ANALYSIS")

print("Q1 =", Q1)

print("Q3 =", Q3)

print("IQR =", IQR)

print("Lower Limit =", lower\_limit)

print("Upper Limit =", upper\_limit)

outliers = df[

(df["Daily\_Sales"] < lower\_limit) |

(df["Daily\_Sales"] > upper\_limit)

]

print("\nOUTLIER RECORDS\n")

print(outliers)

print("\nNumber of Outliers:", len(outliers))

# -----

# 6. Z-Score Outlier Detection

# -----

df["Z\_Score"] = stats.zscore(df["Daily\_Sales"])

```



```

z\_outliers = df[abs(df["Z\_Score"]) > 3]

print("\nZ-SCORE OUTLIERS\n")

print(z\_outliers)

print("\nNumber of Z-Score Outliers:",
len(z\_outliers))

# -----

# 7. Urban and Rural Statistics

# -----

urban = df[df["Branch\_Type"] == "Urban"]["Daily\_Sales"]

rural = df[df["Branch\_Type"] == "Rural"]["Daily\_Sales"]

print("\nURBAN BRANCH STATISTICS")

print(urban.describe())

print("\nRURAL BRANCH STATISTICS")

print(rural.describe())

# -----

# 8. Confidence Interval

# -----

urban\_mean = urban.mean()

urban\_std = urban.std()

urban\_n = len(urban)

```

```

rural\_mean = rural.mean()

rural\_std = rural.std()

rural\_n = len(rural)

urban\_se = urban\_std / np.sqrt(urban\_n)

rural\_se = rural\_std / np.sqrt(rural\_n)

urban\_t = stats.t.ppf(0.975, urban\_n - 1)

rural\_t = stats.t.ppf(0.975, rural\_n - 1)

urban\_ci = (

urban\_mean - urban\_t \* urban\_se,

urban\_mean + urban\_t \* urban\_se

)

rural\_ci = (

rural\_mean - rural\_t \* rural\_se,

rural\_mean + rural\_t \* rural\_se

)

print("\n95% CONFIDENCE INTERVAL")

print("Urban Mean =", urban\_mean)

print("Urban 95% CI =", urban\_ci)

print("Rural Mean =", rural\_mean)

print("Rural 95% CI =", rural\_ci)

```

```

# -----

# 9. Hypothesis Testing

# -----

t_stat, p_two = stats.ttest_ind(

    urban,

    rural,

    equal_var=False

)

# One-tailed test: Urban > Rural

p_one = p_two / 2 if t_stat > 0 else 1 - p_two / 2

print("\nHYPOTHESIS TESTING")

print("Urban Mean =", urban_mean)

print("Rural Mean =", rural_mean)

print("t-statistic =", t_stat)

print("One-tailed p-value =", p_one)

alpha = 0.05

if p_one < alpha:

    print("\nDecision: Reject H0")

    print("Urban branches have significantly higher average sales.")

else:

    print("\nDecision: Fail to reject H0")

```

```
print("There is insufficient evidence that urban branches have higher average sales.")
```

```
# -----
```

```
# 10. Branch Comparison Plot
```

```
# -----
```

```
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(
```

```
data=df,
```

```
x="Branch\Type",
```

```
y="Daily_Sales"
```

```
)
```

```
plt.title("Urban vs Rural Daily Sales")
```

```
plt.xlabel("Branch Type")
```

```
plt.ylabel("Daily Sales")
```

```
plt.show()
```

```
# -----
```

```
# 11. Average Sales by Branch Type
```

```
# -----
```

```
average_sales = df.groupby(
```

```
"Branch\Type"
```

```
)["Daily_Sales"].mean()
```

```
print("\nAVERAGE SALES BY BRANCH TYPE\n")
```

```
print(average\_sales)
```

```
average\_sales.plot(
```

```
kind="bar",
```

```
figsize=(8, 5)
```

```
)
```

```
plt.title("Average Sales: Urban vs Rural")
```

```
plt.xlabel("Branch Type")
```

```
plt.ylabel("Average Daily Sales")
```

```
plt.xticks(rotation=0)
```

```
plt.show()
```

```
# -----
```

```
# 12. Customer Count vs Sales
```

```
# -----
```

```
plt.figure(figsize=(8, 5))
```

```
sns.scatterplot(
```

```
data=df,
```

```
x="Customer\_Count",
```

```
y="Daily\_Sales",
```

```
hue="Branch\_Type"
```

```
)
```

```
plt.title("Customer Count vs Daily Sales")
```

```
plt.xlabel("Customer Count")
```

```
plt.ylabel("Daily Sales")
```

```
plt.show()
```

```
# -----
```

```
# 13. Correlation Analysis
```

```
# -----
```

```
correlation = df[
```

```
[
```

```
"Daily\_Sales",
```

```
"Customer\_Count",
```

```
"Discount\_Percentage",
```

```
"Feedback\_Score"
```

```
]
```

```
].corr()
```

```
print("\nCORRELATION MATRIX\n")
```

```
print(correlation)
```

```
plt.figure(figsize=(8, 6))
```

```
sns.heatmap(
```

```
correlation,
```

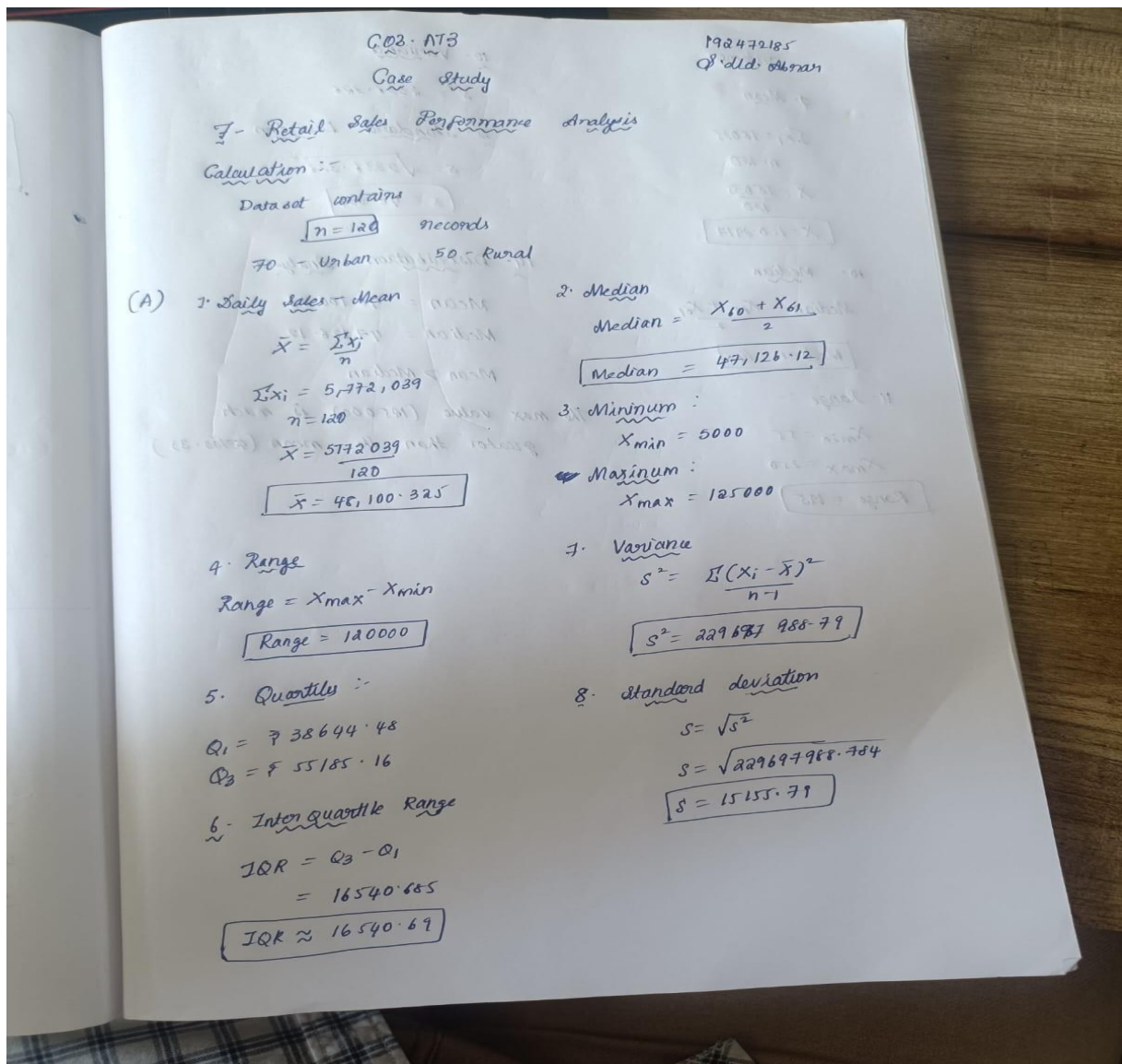
annot=True,

cmap="coolwarm"

)

plt.title("Correlation Matrix")

plt.show()



Customer

9. Mean

$$\sum x_i = 18035$$

$$n = 120$$

$$\bar{X} = \frac{18035}{120}$$

$$\bar{X} = 150.2917$$

10. Median

$$\text{Median} = \frac{X_{60} + X_{61}}{2}$$

$$\text{Median} = 146$$

11. Range :-

$$X_{\min} = 55$$

$$X_{\max} = 250$$

$$\text{Range} = 195$$

B. IQR

$$\text{IQR} = Q_3 - Q_1$$

$$= 55,165,1625 - 38,644,4775$$

$$\text{IQR} = 16540.685$$

$$\text{Lower} = Q_1 - 1.5 (\text{IQR})$$

$$= 13833.45$$

$$\text{Upper} = Q_3 + 1.5 (\text{IQR})$$

$$= 79996.19$$

10. Variance

$$S^2 = 2336.324$$

13. Standard Deviation

$$S = \sqrt{2336.324}$$

$$S = 48.336$$

14. Distribution Analysis

$$\text{Mean} = 48100.33$$

$$\text{Median} = 47126.12$$

$$\text{Mean} > \text{Median}$$

The max value (185000) is much greater than the mean (48100.33)

Zscore Method

$$Z = \frac{X - \bar{X}}{S}$$

Z score for 125000

$$Z = 5.095$$

Z score for 115000

$$Z = 4.433$$

Z score of 5000

$$Z = -2.845$$

$$\text{New Mean} = \frac{5772039 - 245000}{117}$$

$$= 47239.65$$

$$48,100.33 - 47,239.65$$

$$= 860.68$$

Increase by $\nearrow$

C. Urban

$$n_U = 70$$

$$\bar{X}_U = 54,059.87$$

$$S_U = 16,079.40$$

Rural :-

$$n_R = 50$$

$$\bar{X}_R = 39756.96$$

$$S_R = 8484.79$$

$$\text{Difference} :- \bar{X}_U - \bar{X}_R$$

$$= 14302.90$$

$$\frac{54059.87 - 39756.96}{39756.96} \times 100$$

$$= 35.97\%$$

$$CI = \bar{x} \pm t_{d/2, n-1} \frac{s}{\sqrt{n}}$$

$$df = 70 - 1 = 69$$

$$SE = \frac{s}{\sqrt{n}}$$

$$SE = 1921.86$$

$$ME = t \times SE$$

$$ME = 3833.997$$

$$95\% CI_{Urban} = (50,225.87, 57,893.87)$$

$$95\% CI_{Rural} = (37,345.61, 42,168.32)$$

$$\alpha = 0.05$$

Output:

RETAIL SALES DATA

	Date	Branch_Type	Daily_Sales	Customer_Count	Discount_Percentage
\					
0	2026-01-01	Urban	58516.75	181	11.80
1	2026-01-02	Rural	42368.96	170	5.17
2	2026-01-03	Rural	36835.87	111	13.44
3	2026-01-04	Rural	41719.51	108	22.58
4	2026-01-05	Urban	51833.04	151	12.52

	Product_Category	Feedback_Score
0	Grocery	3.85
1	Home Appliances	4.16
2	Grocery	3.83
3	Electronics	2.79
4	Electronics	3.00

DATA INFORMATION

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  120 non-null   object
1   Branch_Type           120 non-null   object
2   Daily_Sales           120 non-null   float64
3   Customer_Count        120 non-null   int64
4   Discount_Percentage   120 non-null   float64
5   Product_Category      120 non-null   object
6   Feedback_Score        120 non-null   float64
dtypes: float64(3), int64(1), object(3)
memory usage: 6.7+ KB
None
```

DESCRIPTIVE STATISTICS

	Daily_Sales	Customer_Count
count	120.000000	120.000000
mean	48100.325000	150.291667
std	15155.790602	48.335556
min	5000.000000	55.000000
25%	38644.477500	114.750000
50%	47126.120000	146.000000
75%	55185.162500	182.750000
max	125000.000000	250.000000

Mean Sales:
48100.325

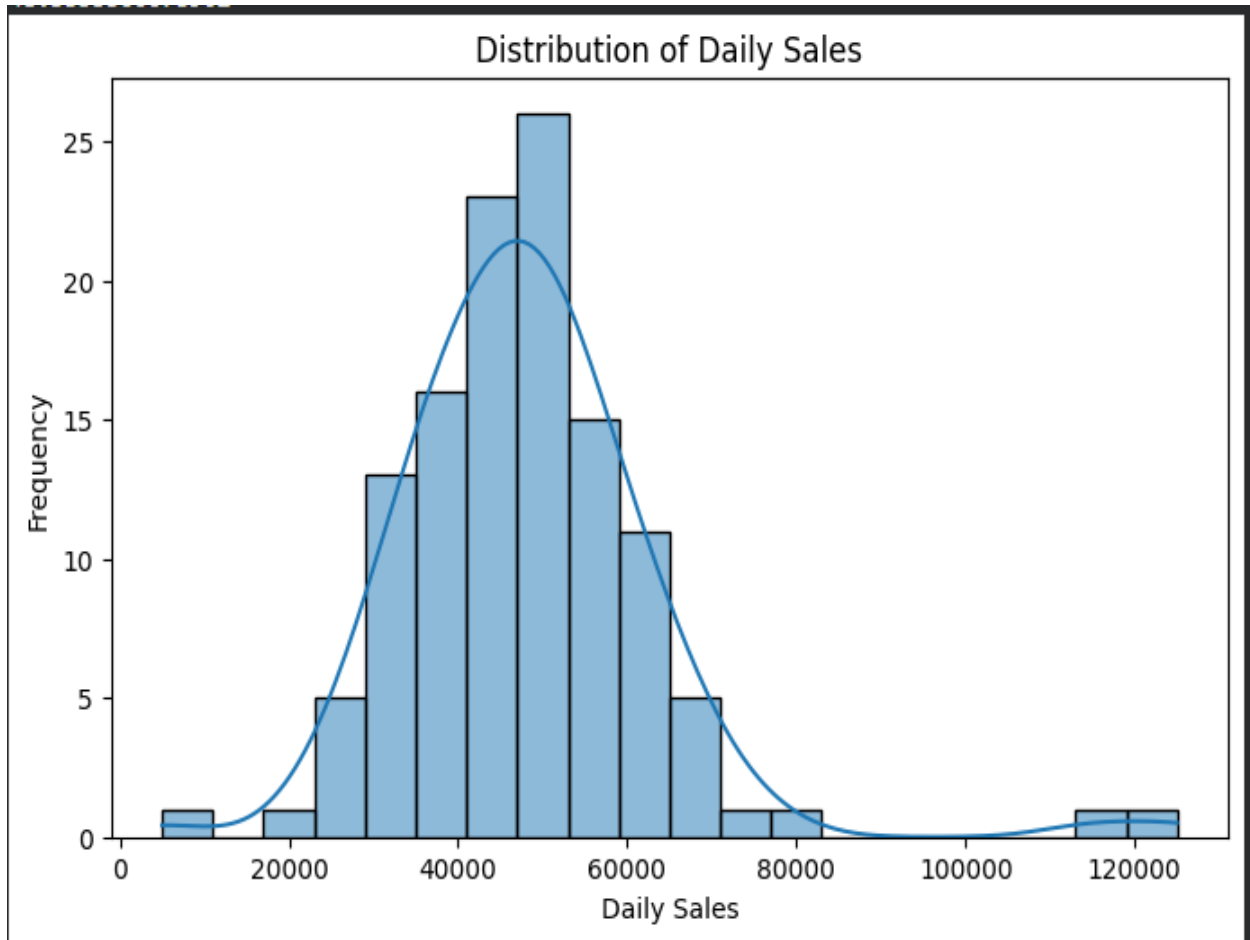
Median Sales:
47126.12

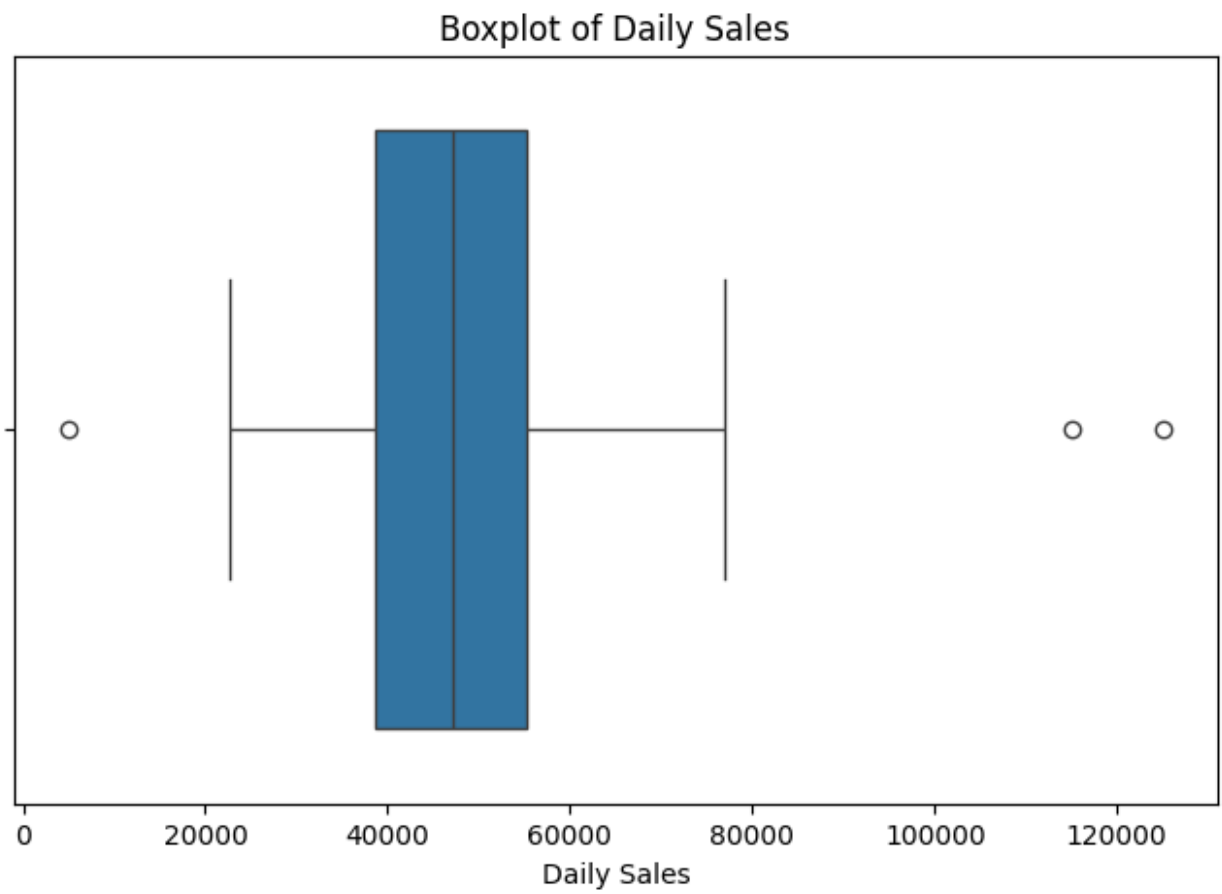
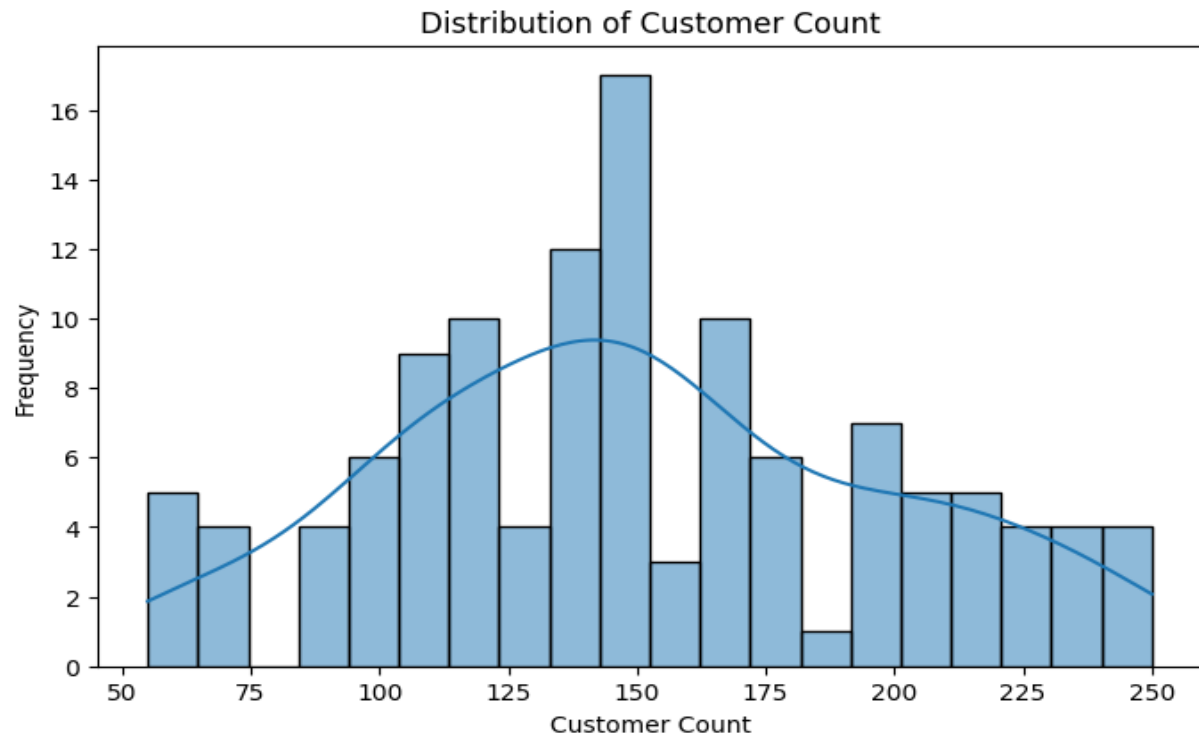
Standard Deviation of Sales:
15155.790602406836

Mean Customer Count:
150.29166666666666

Median Customer Count:
146.0

Standard Deviation of Customer Count:
48.3355560678902





IQR OUTLIER ANALYSIS

Q1 = 38644.4775

Q3 = 55185.1625

IQR = 16540.684999999998

Lower Limit = 13833.450000000004

Upper Limit = 79996.19

OUTLIER RECORDS

	Date	Branch_Type	Daily_Sales	Customer_Count	Discount_Percentage
\					
15	2026-01-16	Urban	125000.0	241	26.27
16	2026-01-17	Urban	115000.0	193	5.13
85	2026-03-27	Urban	5000.0	212	29.20

	Product_Category	Feedback_Score
15	Electronics	3.86
16	Home Appliances	3.57
85	Clothing	3.66

Number of Outliers: 3

Z-SCORE OUTLIERS

	Date	Branch_Type	Daily_Sales	Customer_Count	Discount_Percentage
\					
15	2026-01-16	Urban	125000.0	241	26.27
16	2026-01-17	Urban	115000.0	193	5.13

	Product_Category	Feedback_Score	Z_Score
15	Electronics	3.86	5.095221
16	Home Appliances	3.57	4.432641

Number of Z-Score Outliers: 2

URBAN BRANCH STATISTICS

```
count      70.000000
mean      54059.868714
std       16079.400587
min        5000.000000
25%       46826.212500
50%       52386.550000
75%       60097.225000
max      125000.000000
Name: Daily_Sales, dtype: float64
```

RURAL BRANCH STATISTICS

```
count      50.0000
mean      39756.9638
std       8484.7932
min      22600.9400
25%      33703.2300
50%      40972.1550
75%      45055.7525
max      57151.5500
Name: Daily_Sales, dtype: float64
```

95% CONFIDENCE INTERVAL

Urban Mean = 54059.86871428571

Urban 95% CI = (np.float64(50225.87097789525), np.float64(57893.86645067617))

Rural Mean = 39756.9638

Rural 95% CI = (np.float64(37345.612253062616),
np.float64(42168.31534693738))

HYPOTHESIS TESTING

Urban Mean = 54059.86871428571

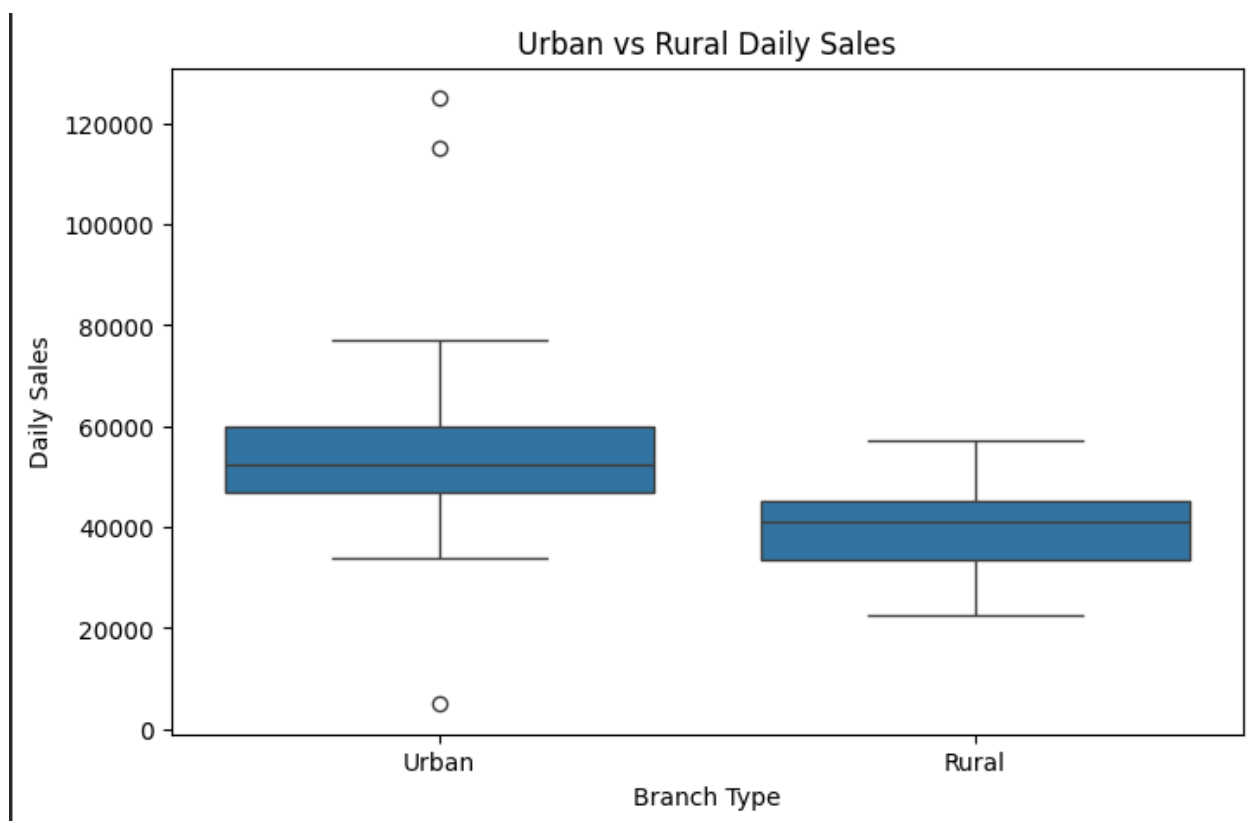
Rural Mean = 39756.9638

t-statistic = 6.312816912534508

One-tailed p-value = 2.978704821216235e-09

Decision: Reject H0

Urban branches have significantly higher average sales.



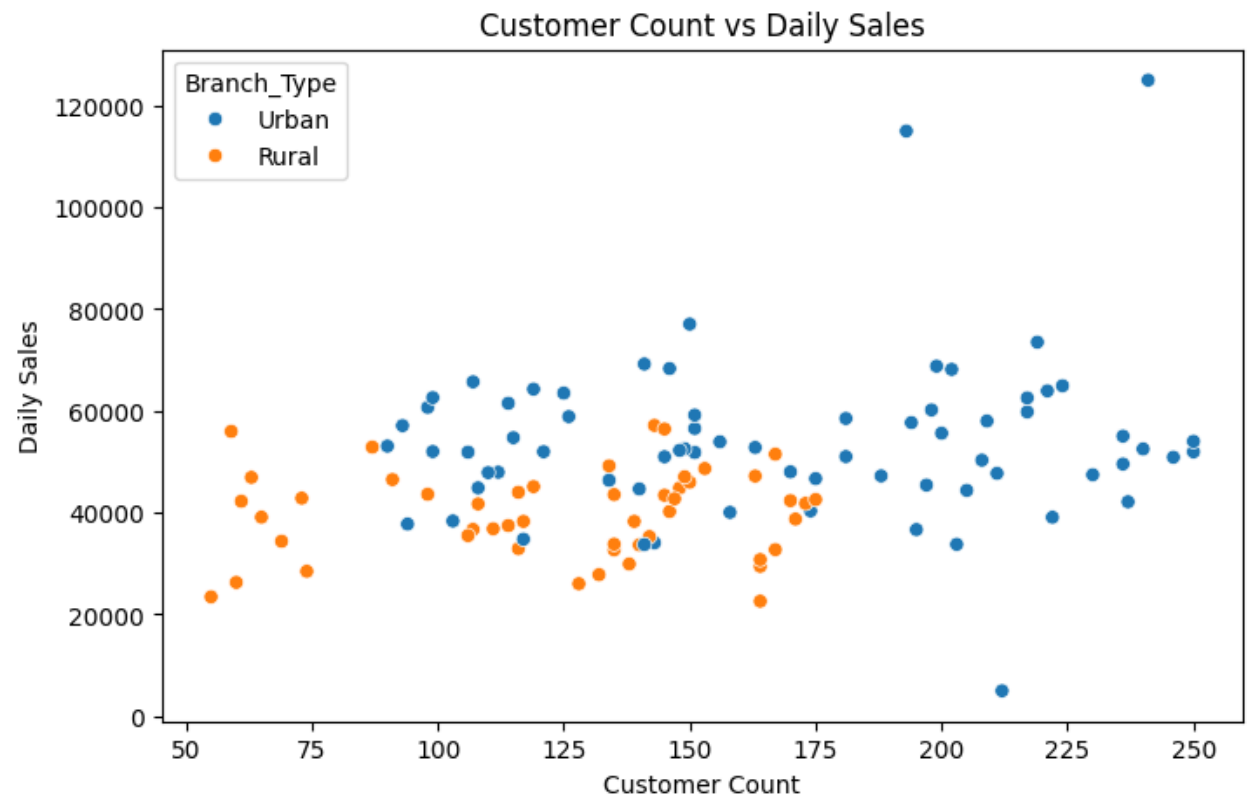
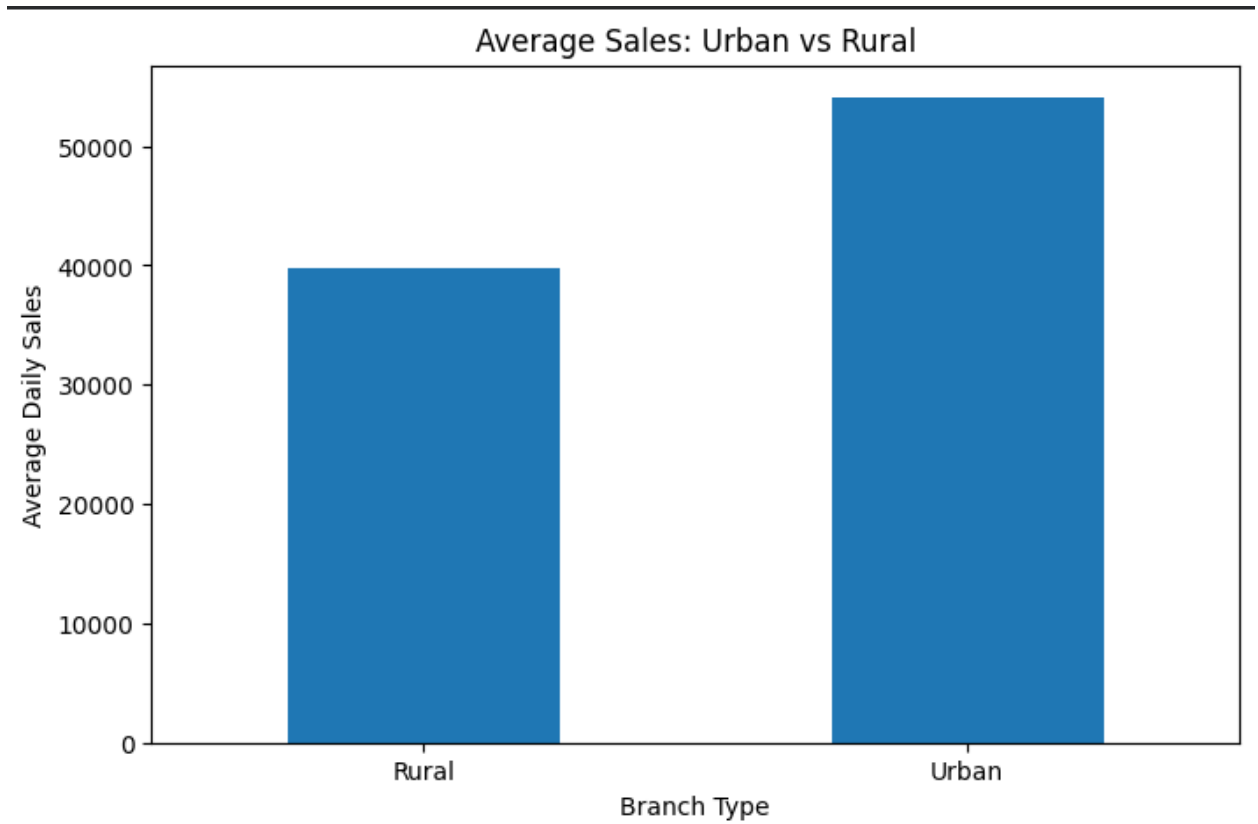
AVERAGE SALES BY BRANCH TYPE

Branch_Type

Rural 39756.963800

Urban 54059.868714

Name: Daily_Sales, dtype: float64



CORRELATION MATRIX

	Daily_Sales	Customer_Count	Discount_Percentage	\
Daily_Sales	1.000000	0.283452	-0.096341	
Customer_Count	0.283452	1.000000	-0.021147	
Discount_Percentage	-0.096341	-0.021147	1.000000	
Feedback_Score	-0.093913	-0.145525	-0.024536	

	Feedback_Score
Daily_Sales	-0.093913
Customer_Count	-0.145525
Discount_Percentage	-0.024536
Feedback_Score	1.000000

